10.4 Configuring an application for multiple environments

In this section you'll learn about hosting environments in ASP.NET Core. You'll learn how to set and determine which environment an application is running in and how to change which configuration values are used, based on the environment. Using environments lets you switch easily among different sets of configuration values in production compared with development, for example.

Any application that makes it to production will likely have to run in multiple environments. If you're building an application with database access, for example, you'll probably have a small database running on your machine that you use for development. In production, you'll have a completely different database running on a server somewhere else.

Another common requirement is to have different amounts of logging depending on where your app is running. In development, it's great to generate lots of logs, which help with debugging, but when you get to production, too many logs can be overwhelming. You'll want to log warnings, errors, and maybe information-level logs, but definitely not debug-level logs!

To handle these requirements, you need to make sure that your app loads different configuration values depending on the environment it's running in: load the production database connection string when in production, and so on. You need to consider three aspects:

- How your app identifies the environment it's running in
- How you load different configuration values based on the current environment
- How to change the environment for a particular machine

This section tackles these aspects in turn so that you can easily tell your development machine apart from your production servers and act accordingly.